

SOP-PV-300: Pressure Vessel Inspection Procedure

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Section 1. Purpose and Scope

This procedure defines the internal and external inspection requirements for pressure vessels operating above 15 psig, in accordance with API 510 and site risk-based inspection (RBI) policy. It covers storage vessels, separators, reactors, and heat exchangers classified as pressure vessels under the site equipment register.

Section 2. Inspection Intervals

Clause 2.1 — External visual inspection shall be performed every 5 years, or per the interval calculated by the site RBI program, whichever is shorter.

Clause 2.2 — Internal inspection (requiring vessel entry and de-inventory) shall be performed at intervals not exceeding 10 years, or half the remaining corrosion-allowance life as calculated by the Mechanical Integrity Engineer, whichever is shorter.

Clause 2.3 — Vessels in known corrosive or cyclic service shall follow the accelerated inspection interval assigned by the RBI risk ranking, which may be as frequent as every 3 years.

Clause 2.4 — Thickness monitoring locations (TMLs) shall be measured using ultrasonic testing (UT) at each scheduled inspection, and corrosion rates shall be recalculated and compared against the previous three inspection cycles.

Section 3. External Inspection Procedure

Step 3.1 — Inspect external surfaces for coating breakdown, corrosion under insulation (CUI), and mechanical damage. Insulation shall be removed at representative locations identified by the CUI risk assessment.

Step 3.2 — Inspect all nozzles, manways, and support skirts for cracking, corrosion, or foundation settlement exceeding 0.5 degrees of tilt.

Step 3.3 — Verify the pressure relief device is correctly installed, in date for its recertification interval per SOP-VLV-100 Section 2, and set at the correct pressure per the vessel nameplate.

Step 3.4 — Document all findings on Form MI-021 (External Vessel Inspection Report), including photographs of any anomalies rated Moderate or Severe.

Section 4. Internal Inspection Procedure

Clause 4.1 — Before internal inspection, the vessel shall be isolated, drained, purged, and confined-space entry procedures per SOP-CSE-200 shall be followed in full, including atmospheric testing and permit issuance.

Clause 4.2 — Internal surfaces shall be inspected for pitting, cracking, weld defects, and general corrosion. Minimum remaining wall thickness shall be compared against the calculated minimum required thickness per ASME Section VIII.

Clause 4.3 — Any indication of cracking shall be evaluated by a qualified inspector using magnetic particle testing (MT) or dye penetrant testing (PT), and the vessel shall not be returned to service until a fitness-for-service assessment is complete.

Clause 4.4 — If measured wall thickness at any TML is within 10 percent of the calculated minimum required thickness, the vessel shall be flagged for engineering review and the inspection interval shall be shortened per Clause 2.3.

Section 5. Fitness-for-Service and Repair

Step 5.1 — Vessels found with corrosion or defects exceeding acceptance criteria shall undergo a fitness-for-service evaluation per API 579 before continued operation is approved.

Step 5.2 — Repairs shall be performed per the site Welding and Repair Procedure and shall be re-inspected and re-tested before the vessel is returned to service.

Step 5.3 — Any repair affecting the pressure boundary requires sign-off from the Mechanical Integrity Engineer and, where required by jurisdiction, a regulatory inspector.

Section 6. Records and Reporting

Clause 6.1 — All inspection reports, UT thickness readings, and fitness-for-service evaluations shall be retained in the Vessel Integrity File for the operating life of the vessel plus 10 years.

Clause 6.2 — Any vessel condition that could result in loss of containment shall be reported to the Process Safety Engineer within 24 hours, independent of the standard reporting cycle described in Section 3.